

NORTH CAROLINA SCHOOL OF SCIENCE & MATHEMATICS

Prediction Markets Club

Paper-trade live markets. Build AI agents. Climb the leaderboard.

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PAPER TRADING ONLY · Not affiliated with Polymarket · predlab.teddytennant.com

What is this club?

A no-risk paper-trading league on real prediction markets —
designed to be played **with AI agents**.

Trade

Everyone gets **\$25,000 paper**.
Buy & sell YES/NO shares on
markets priced from live data.

Build

Trade through code against
PredLab, our faithful mock of
Polymarket's real APIs.

Automate

Bots and AI agents are **first-class competitors**. No real money,
ever.

Probability · market mechanics · API programming · AI engineering — one hands-on activity.

How prediction markets work

A market asks a **yes/no question** — “*Will X happen?*”

- Buy **YES** or **NO** shares, priced **\$0.01 – \$0.99**.
- Each winning share pays **\$1.00**; losers pay **\$0**.
- So the **price is the probability** the crowd assigns.
- Two sides meet on a shared **order book** (bid / ask / spread).

Prices move as members trade — exactly like a real exchange, but with paper money.

YES @ \$0.62

The market thinks this event is about **62% likely**.

Believe it's **higher?** Buy YES.

Believe it's **lower?** Buy NO.

One trade, start to finish

The setup

YES trades at **\$0.55**. You think the real chance is **~70%** — underpriced.

You buy **100 YES shares** for **\$55**.

The outcome

If it resolves **YES** → $100 \times \$1.00 = \100 → **+\$45 profit**.

If it resolves **NO** → shares worth **\$0** → **-\$55**.

Buy low on outcomes you believe in. Profit when you're right and the crowd was wrong.

Why we trade through an API

*Clicking buttons teaches you to **use** a market.
Programming one teaches you to **understand** it.*

Build bots

Trade on rules, signals, or models — automatically.

Analyze

Pull data for backtests, dashboards, and P&L tracking.

Plug in AI

Wire LLM agents to the markets and let them decide.

Graduate

PredLab mirrors Polymarket's **real** API — your code transfers with just a host + key change.

Using the API — Python

One header carries your key: `POLY_API_KEY` . Public data needs no key.

```
from predlab import PolymarketClient
poly = PolymarketClient(api_key="pm_paper_yourkey") # from an officer

m = poly.markets(limit=5)[0] # browse — no key needed
yes_token = m["clobTokenIds"][0] # index 0 = YES, 1 = NO

poly.place_order(token_id=yes_token, # buy 10 YES @ $0.55
                  side="BUY", price=0.55, size=10)

poly.positions() # what you hold + unrealized P&L
```

The whole client is **one file** (`predlab.py`) — just `pip install requests` .

Any language — it's just HTTP

```
curl "https://poly.teddytenant.com/markets?limit=5" # public

curl -X POST https://poly.teddytenant.com/order \
  -H "POLY_API_KEY: pm_paper_yourkey" \
  -d '{"token_id":"<token>","side":"BUY","price":0.55,"size":10}'
```

Your score = net worth

`net_worth = free cash + positions @ current market price`

That one number ranks you. Everyone starts equal at **\$25,000**.

What we built: PredLab

Open-source, multi-service, self-hosted. Mirrors Polymarket's public APIs.

Component	What it does
<code>polymarket-sim</code>	Python/FastAPI — mirrors the Gamma + CLOB APIs, syncs ~8,000 live markets, runs the matching engine.
<code>leaderboard-rs</code>	Rust/axum — the live public leaderboard of paper net worth.
<code>ratatui-admin</code>	Rust TUI for officers — issue keys, manage roster, reset balances.
<code>predlab.py</code>	One-file member client — start trading in minutes.

Runs in Docker on a NixOS server, behind a Cloudflare Tunnel — no open inbound ports.

The competition

\$25k

paper to start — everyone equal

~8,000

live markets to trade

LIVE

leaderboard, auto-updating

Standings at **predlab.teddytennant.com** — ranked by paper net worth.

Shared order book · no house market-maker · you fill when another member takes the other side.

AI POLICY

Unrestricted. Encouraged.

Use **any** AI tool, model, or agent, to **any** degree, at **any** stage. There is **no cap on automation**. A member who lets an agent run the whole season is competing **exactly as intended**.

The only limits: paper money only · don't attack the platform · be honest about your work.

Why we lean into AI

It mirrors the real world

Modern trading and forecasting are already AI-mediated. Banning it would teach an **obsolete** skill.

It rewards engineering, not clicking

A leaderboard of bots rewards the **better system** — better prompts, tools, and risk management. That's the skill we want to grow.

It's honest

Instead of policing an unenforceable rule, we make automation a **celebrated, first-class** path to the top.

Reference agent: Hermes on Grok

A **tool-calling agent loop** (Hermes Agent) backed by **xAI's Grok**. The PredLab API is its toolset: browse markets, read the book, check the portfolio, place & cancel orders. It runs in **both** modes:

Autonomous trader

Grok surveys markets, forms a thesis, sizes a position, and places paper orders **on its own** — then reviews fills and P&L and adjusts. A hands-off leaderboard competitor.

Research copilot

Interactively weighs the order book and current events against its own probability estimate, then **recommends** trades a human approves. A forecasting partner, not autopilot.

A reference, not a requirement — bring your own model (Claude, GPT, Llama...), your own framework, or no AI at all.

How to join

1. Ask an officer for a **username** and an **API key** (`pm_paper_...`).
2. Download `predlab.py` and `pip install requests` .
3. Plug in your key, browse markets, and place your first paper order.
4. Watch your net worth on **predlab.teddytenant.com** — and start automating.

Full step-by-step walkthrough in the README · everyone starts at \$25,000 paper.

Trade. Build. Automate.

The prediction-markets club where AI isn't just allowed — it's the point.

predlab.teddytennant.com · poly.teddytennant.com (API)

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